

Cryptography 1 Project Submission



These are the expectations for the final project for this course

Posted Tue Sep 15, 2026 at 2:43 pm

Criteria	Grading Scale	
Demonstrate Cryptographic Fundamentals The project must include a report demonstrating an understanding of fundamental cryptography concepts. The report must include the implementation of 2 basic cryptographic primitives using Python, documentation of security properties for each implementation, a clear explanation of the mathematical principles used, and an analysis of security guarantees provided by the implementations.	1 Complete	0 Incomplete
Implement Symmetric Cryptography The project must include the creation and documentation of a symmetric encryption implementation. The implementation must include a working example of AES encryption in both ECB and CBC modes, a demonstration of proper key management techniques, the implementation of 1 stream cipher, and documentation of security considerations and best practices	1 Complete	0 Incomplete
Apply Hash Functions and Integrity The project must include the development and documentation of hash function implementations, including the creation of a secure password hashing system using modern algorithms, the implementation of HMAC for message authentication, a file integrity verification system, and documentation of implementation security properties.	1 Complete	0 Incomplete
Utilize Public Key Cryptography The project must include documentation demonstrating the generation and management of RSA key pairs, the implementation of basic asymmetric encryption, the setup of a simple PKI environment, and a demonstration of secure key exchange	1 Complete	0 Incomplete
Implement Digital Signatures The project must include the creation of a digital signature system, including the implementation of RSA-based digital signatures, certificate generation and management, a signature verification process, and documentation of the trust model implementation.	1 Complete	0 Incomplete
Configure Cryptographic Protocols The project must include the demonstration of protocol implementation, including the configuration of TLS for a web server, the setup of SSH with key-based authentication, the implementation of a secure communication channel, and an analysis of protocol security using Wireshark	1 Complete	0 Incomplete

Total pts: 6